Project Writeup: Food Access and Adult Obesity in Chicago

DSC 510, Health Data Science | *Group 8* | *Heema Shah & Anas Kapadia*

What I Worked On

For this project I handled the data side: pulling the two datasets together, cleaning them, and running the analysis. Here is how I did it and what I found.

Getting the Data Ready

I used two datasets. One was the CDC PLACES dataset, which gives adult obesity rates for every census tract. The other was the USDA Food Access Research Atlas, which tells you whether a tract is classified as a food desert (LILA, meaning low-income and low-access).

Both datasets used Census Tract FIPS codes, so I used those to match the two files together. Before I could do that, I had to convert the FIPS codes into a consistent numeric format, since they were not formatted the same way across both files. Once that was done, I used XLOOKUP in Excel to pull the obesity rate into the food access dataset for each matching tract.

The CDC dataset had a lot of health indicators in it, not just obesity, so I filtered it down to only the obesity numbers. After merging, I went through and removed any rows that had missing values, since those would throw off the statistics. What I was left with was a clean dataset of 840 Chicago census tracts, each with an obesity rate and a food desert classification.

Running the Analysis

Once the data was clean, I ran some basic descriptive statistics on the obesity rates to see the overall spread. The average came out to 25.23%, but the range was huge, from 1.4% up to 92.2%, which told me obesity is not spread evenly across Chicago at all. Some neighborhoods are dealing with a much bigger burden than others.

Then I ran a linear regression to actually test the relationship: does living in a food desert predict a higher obesity rate? I set it up with obesity rate as the outcome and food desert status (0 for high access, 1 for food desert) as the predictor.

The regression came back with a coefficient of 8.307 and a p-value of 0.015. That means tracts classified as food deserts had obesity rates about 8.3 percentage points higher than tracts with better access, and since the p-value is under 0.05, that result is statistically significant. It is not just noise.

The R^2 was low, 0.007, which I expected. Obesity is affected by a lot more than just food access alone, things like income, physical activity, education, and healthcare access all play a role. So food access on its own only explains a small slice of the picture. But the fact that it still came out statistically significant tells me it is a real, measurable factor, not something to ignore.

Putting It Into a Visual

To show the relationship clearly, I built a scatter plot with the regression line fitted on top of it, plotting food desert status against obesity rate. The upward slope on the line makes the pattern easy to see at a glance: food desert tracts cluster higher on obesity than non-food-desert tracts.

What This Means

The takeaway for me is that food access is a real, quantifiable piece of why obesity rates differ so much across Chicago neighborhoods. It is not the whole story, but it is significant enough that it is worth acting on. That is part of why our group proposed community-level interventions, like supporting corner stores to carry fresh produce and running mobile produce markets, targeted at the tracts this analysis flagged as highest risk.

Tools used: Microsoft Excel (XLOOKUP, descriptive statistics, linear regression, scatter plot with trendline)